

AP Statistics MCQ Practice Exam 05

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AP Statistics MCQ Practice Exam 5

PART A, QUESTIONS

APSTAT-MCQ5-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Six temperature deviations from their sample mean are -4 , -1 , 2 , 3 , 5 , and k degrees Celsius. What must k equal?

- A) -5
- B) 0
- C) 5
- D) 15

APSTAT-MCQ5-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

An opt-in website poll of 40,000 visitors is used to test a claim about the proportion of all city residents who favor a zoning change. The test gives $p\text{-value} < 0.0001$. Which assessment is most appropriate?

- A) The tiny p -value guarantees that the poll represents all city residents.
- B) The result proves the population proportion differs from the null value by a practically important amount.
- C) The tiny p -value does not repair possible voluntary-response and undercoverage bias in the poll.
- D) The large number of responses converts the opt-in poll into a simple random sample.

APSTAT-MCQ5-003

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

A laboratory selects one random sample of rechargeable lamps and wants to compare their population mean operating time with a claimed value of 9.5 hours. Which procedure matches the research goal?

- A) A two-proportion z-test
- B) A two-sample t-test
- C) A paired t-test
- D) A one-sample t-test for a population mean

APSTAT-MCQ5-004

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For six paired observations, the sum of the products of their standardized scores is $\sum(z_x z_y) = -3.5$. What is the correlation r ?

- A) -0.58
- B) -0.70
- C) 0.70
- D) -3.50

APSTAT-MCQ5-005

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ5-005 through APSTAT-MCQ5-007 use the following stimulus. Let X be the number of optional security features included in a randomly selected smart-home installation. Possible values are $x = 0, 1, 2, 3$, and the probability model is $P(X = x) = c(x + 1)$, where c is a constant.

What value of c makes this a valid probability distribution?

- A) 0.40
- B) 0.25
- C) 0.20
- D) 0.10

APSTAT-MCQ5-006

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ5-005 through APSTAT-MCQ5-007 use the following stimulus. Let X be the number of optional security features included in a randomly selected smart-home installation. Possible values are $x = 0, 1, 2, 3$, and the probability model is $P(X = x) = c(x + 1)$, where c is a constant.

What is $P(X \geq 2)$?

- A) 0.70
- B) 0.50
- C) 0.40
- D) 0.30

APSTAT-MCQ5-007

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ5-005 through APSTAT-MCQ5-007 use the following stimulus. Let X be the number of optional security features included in a randomly selected smart-home installation. Possible values are $x = 0, 1, 2, 3$, and the probability model is $P(X = x) = c(x + 1)$, where c is a constant.

Which statement correctly describes the most likely value of X ?

- A) $X = 0$ is most likely because it is the smallest possible count.
- B) $X = 3$ is most likely, with probability 0.40, but an individual installation is not guaranteed to have 3 features.
- C) $X = 2$ is most likely because $P(X \geq 2)$ is greater than 0.50.
- D) Every group of 10 installations contains exactly 4 with $X = 3$.

APSTAT-MCQ5-008

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A bottling line fills products in a repeating 12-position mechanical cycle. An inspector chooses a random starting bottle and then inspects every 12th bottle. What is the main concern with this systematic sample?

- A) It contains too many bottles to be considered a sample.
- B) A random start makes periodic production patterns irrelevant.
- C) The sampling interval matches the production cycle, so the sample may repeatedly observe the same cycle position.
- D) Systematic sampling requires inspecting every bottle in the first cycle.

APSTAT-MCQ5-009

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Suppose 200 independent significance tests are conducted at $\alpha = 0.05$ and every null hypothesis is actually true. What is the expected number of Type I errors?

- A) 10
- B) 4
- C) 20
- D) 190

APSTAT-MCQ5-010

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For a right-tailed one-sample t-test of $H_0: \mu = 50$, the standard error is 2 and the critical value at the chosen significance level is 1.711. For what sample mean would the test just begin to reject H_0 ?

- A) $\bar{x} = 50.856$
- B) $\bar{x} = 51.711$
- C) $\bar{x} = 52.000$
- D) $\bar{x} = 53.422$

APSTAT-MCQ5-011

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

An engineer wants to estimate $P(\text{overheats} \mid \text{fast-charged})$ for a battery model. Which data-collection plan most directly targets this conditional probability?

- A) Select only batteries that overheated and determine how many were fast-charged.
- B) Randomly select fast-charged batteries and record the proportion that overheat.
- C) Select equal numbers of overheated and non-overheated batteries, regardless of charging method.
- D) Multiply the overall overheating rate by the overall fast-charging rate.

APSTAT-MCQ5-012

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

In a density histogram, the bar covering values from 10 up to 15 has height 0.04. What proportion of observations is represented by that bar?

- A) 0.04
- B) 0.09
- C) 0.20
- D) 0.50

APSTAT-MCQ5-013

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

Which research question is most naturally addressed with a least-squares regression analysis?

- A) What proportion of vehicles pass an annual safety inspection?
- B) Do two fertilizers produce different mean crop yields?
- C) How is engine temperature associated with fuel efficiency across a sample of vehicles?
- D) Does the distribution of preferred colors differ among age groups?

APSTAT-MCQ5-014

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A city wants to estimate the proportion of residents who maintain an emergency supply kit. Which sampling frame would most directly reduce the undercoverage created by using only a published telephone directory?

- A) A probability sample of residential addresses, including households without listed telephone numbers
- B) A poll posted on the city's emergency-management social media page
- C) A sample of customers leaving outdoor-supply stores
- D) A telephone-directory sample enlarged until it contains 5,000 households

APSTAT-MCQ5-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Independent random samples compare mean drying time for coating A and coating B. The estimate $\bar{x}_A - \bar{x}_B$ is 4.6 minutes with standard error 1.8 minutes. For $df = 38$, use $t^* = 2.024$ for a 95% confidence interval. Which interval is correct?

- A) (2.80, 6.40) minutes
- B) (0.96, 8.24) minutes
- C) (-3.64, 3.64) minutes
- D) (-0.04, 9.24) minutes

APSTAT-MCQ5-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A student's score is the 28th value when 36 scores are ordered from smallest to largest. Approximately what percentile is the score, using the percentage of observations at or below it?

- A) 22nd percentile
- B) 36th percentile
- C) 72nd percentile

D) 78th percentile

APSTAT-MCQ5-017

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Let X be a service cost with $\mu_X = \$80$ and $\sigma_X = \$20$. A total charge is defined by $Y = 1.08X + 5$, where 1.08 applies a percentage surcharge and \$5 is a fixed fee. What are μ_Y and σ_Y ?

A) $\mu_Y = \$91.40$ and $\sigma_Y = \$21.60$

B) $\mu_Y = \$91.40$ and $\sigma_Y = \$26.60$

C) $\mu_Y = \$86.40$ and $\sigma_Y = \$21.60$

D) $\mu_Y = \$91.40$ and $\sigma_Y = \$25.00$

APSTAT-MCQ5-018

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A workplace study asks, "Among all employees who work remotely at least three days per week, what proportion report feeling professionally isolated?" Which parameter matches the question?

A) The proportion of all employees who both work remotely and report isolation

B) The sample proportion of surveyed remote employees who report isolation

C) The population proportion p of remote employees who report isolation

D) The population proportion of isolated employees who work remotely

APSTAT-MCQ5-019

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

An ecologist wants to estimate mean soil moisture across a preserve containing meadow, woodland, and marsh habitat. Moisture varies strongly by habitat. Which plan is most appropriate?

A) Sample only the habitat with the largest area.

B) Divide the preserve by habitat and randomly sample locations within every habitat.

- C) Select a few convenient points along the preserve's main trail.
- D) Randomly select one habitat and measure every location in it.

APSTAT-MCQ5-020

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

For repeated random samples of the same size, the standard error of $\bar{x}$ is 0.8 milligram. Which interpretation is correct?

- A) Every observation lies within 0.8 milligram of the population mean.
- B) The population standard deviation is necessarily 0.8 milligram.
- C) Eighty percent of sample means lie above the population mean.
- D) The sample mean typically varies by about 0.8 milligram from sample to sample around μ .

APSTAT-MCQ5-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For a linear relationship between soil salinity and seedling height, $r = -0.45$. Approximately what percentage of the variation in seedling height is not explained by the least-squares linear relationship with salinity?

- A) 20.25%
- B) 45.00%
- C) 79.75%
- D) 95.50%

APSTAT-MCQ5-022

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

Which question is represented by the conditional probability $P(\text{late delivery} \mid \text{expedited order})$?

- A) Among expedited orders, what proportion are delivered late?

- B) Among late deliveries, what proportion were expedited?
- C) What proportion of all orders are both late and expedited?
- D) Are late delivery and expedited status mutually exclusive?

APSTAT-MCQ5-023

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A simple random sample of 50 customers contains 2 who request a paper receipt. Why is a standard one-proportion z-confidence interval questionable?

- A) The sample proportion is less than 0.50.
- B) The sample size is greater than 30.
- C) The sample contains more failures than successes.
- D) The sample has fewer than 10 successes.

APSTAT-MCQ5-024

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Two laboratories measure particle diameters using the same histogram bins. Their relative-frequency histograms are identical, but laboratory A measured 80 particles and laboratory B measured 200 particles. Which statement must be true?

- A) Each corresponding bin contains the same number of particles in both laboratories.
- B) Each corresponding bin contains the same proportion of its laboratory's sample, though the raw bin counts can differ.
- C) The two samples have exactly the same mean and standard deviation.
- D) The two laboratories measured the same individual particles.

APSTAT-MCQ5-025

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

Two sensors will measure temperature at a shaded site and a sunny site. Sensor 1 tends to read slightly higher than sensor 2. Which design best prevents sensor bias from being confounded with site?

- A) Randomly rotate the sensors between the two sites across measurement periods and record sensor and site for each reading.
- B) Place sensor 1 permanently at the sunny site and sensor 2 permanently at the shaded site.
- C) Use only sensor 1 at both sites, but measure the sunny site first every day.
- D) Let the technician choose a sensor for each site based on the day's forecast.

APSTAT-MCQ5-026

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

At an airport, the probabilities that a flight is delayed specifically by weather or specifically by a mechanical issue are 0.12 and 0.05. These two listed causes are mutually exclusive for the reporting system. What is the probability that a flight is delayed by either of these causes?

- A) 0.006
- B) 0.07
- C) 0.17
- D) 0.60

APSTAT-MCQ5-027

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A random sample finds a statistically significant association between vegetarian diet status and supplement use in a population. Diet was not randomly assigned. Which conclusion is justified?

- A) Vegetarian diets cause increased supplement use because the sample was random.
- B) The association may be generalized to the sampled population if the sampling design was sound, but causation is not established.
- C) No population conclusion is possible because random assignment was not used.
- D) Statistical significance eliminates the possibility of confounding.

APSTAT-MCQ5-028

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A researcher wants to study “engagement” during collaborative writing sessions. Which operational definition gives a clear quantitative variable for each participant?

- A) Whether the participant seemed engaged to the researcher
- B) The participant's favorite writing topic
- C) Whether the group produced a good document
- D) The number of minutes the participant actively edited the shared document during the session

APSTAT-MCQ5-029

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A confidence interval for $p_A - p_B$ is $(-0.04, 0.12)$. What was the observed point estimate $\hat{p}_A - \hat{p}_B$ at the center of the interval?

- A) -0.08
- B) 0.02
- C) 0.04
- D) 0.08

APSTAT-MCQ5-030

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ5-030 through APSTAT-MCQ5-032 use the following stimulus. For a sample of green roofs, x is soil depth in centimeters and y is midday summer surface temperature in degrees Celsius. The least-squares line is $\hat{y} = 44.6 - 0.38x$. Analysts define a centered explanatory variable $u = x - 15$.

Which equation expresses the same prediction line using u instead of x ?

- A) $\hat{y} = 38.9 - 0.38u$
- B) $\hat{y} = 44.6 - 0.38u$

C) $\hat{y} = 50.3 - 0.38u$

D) $\hat{y} = 38.9 + 0.38u$

APSTAT-MCQ5-031

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ5-030 through APSTAT-MCQ5-032 use the following stimulus. For a sample of green roofs, x is soil depth in centimeters and y is midday summer surface temperature in degrees Celsius. The least-squares line is $\hat{y} = 44.6 - 0.38x$. Analysts define a centered explanatory variable $u = x - 15$.

What is the interpretation of the intercept in the regression equation written in terms of u ?

- A) It is the predicted temperature decrease for each additional 15 centimeters of soil.
- B) It is the observed mean temperature for every roof in the sample.
- C) It is the predicted temperature at an impossible soil depth of 0 centimeters.
- D) It is the predicted surface temperature, 38.9°C , for a roof with soil depth 15 centimeters.

APSTAT-MCQ5-032

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ5-030 through APSTAT-MCQ5-032 use the following stimulus. For a sample of green roofs, x is soil depth in centimeters and y is midday summer surface temperature in degrees Celsius. The least-squares line is $\hat{y} = 44.6 - 0.38x$. Analysts define a centered explanatory variable $u = x - 15$.

For a least-squares line with an intercept, which statement about the residuals for the fitted sample is correct, apart from minor rounding?

- A) Every residual equals zero.
- B) The residuals sum to zero, so positive and negative prediction errors balance algebraically.
- C) The residuals all have the same absolute value.
- D) The residuals sum to the intercept.

APSTAT-MCQ5-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Before distributing a statewide questionnaire, researchers administer it to 25 people similar to the target population and interview them about how they understood each item. What is the main purpose of this pilot test?

- A) To identify ambiguous wording and response-option problems before the main study
- B) To replace the need for a probability sample in the main study
- C) To guarantee that no respondent in the main study will skip a question
- D) To produce the final statewide estimates from the 25 participants

APSTAT-MCQ5-034

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

In 2,000 independent simulation trials, an event occurs 620 times, giving $\hat{p} = 0.31$. Using the simulated trials as Bernoulli observations, what is the approximate standard deviation of this simulated proportion?

- A) 0.1034
- B) 0.0214
- C) 0.0103
- D) 0.0001

APSTAT-MCQ5-035

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A two-sided t-test has test statistic $t = 1.90$ with $df = 20$. The critical values at $\alpha = 0.05$ are ± 2.086 . Which decision is correct?

- A) Reject H_0 because 1.90 is positive.
- B) Reject H_0 because 1.90 is greater than 0.05.
- C) Fail to reject H_0 because df is less than 30.
- D) Fail to reject H_0 because $|1.90| < 2.086$.

APSTAT-MCQ5-036

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A researcher will compare two population proportions using independent samples with a fixed total of 400 observations. If the two population proportions are expected to be similar, which allocation generally gives the smallest standard error for the estimated difference?

- A) 350 observations in sample 1 and 50 in sample 2
- B) 200 observations in each sample
- C) 300 observations in sample 1 and 100 in sample 2
- D) Any allocation gives the same standard error when the total is fixed

APSTAT-MCQ5-037

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Every value in a quantitative dataset is converted to a sample z-score using $z = (x - \bar{x})/s$. What are the sample mean and sample standard deviation of the resulting z-scores?

- A) Mean 0 and standard deviation 1
- B) Mean 1 and standard deviation 0
- C) Mean $\bar{x}$ and standard deviation s
- D) Mean 0 and standard deviation s^2

APSTAT-MCQ5-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For a binomial random variable X , $\mu_X = 12$ and $\sigma_X = 3$. Which interpretation of σ_X is appropriate?

- A) X can never differ from 12 by more than 3.
- B) Exactly 68% of X -values are between 9 and 15 without any additional condition.
- C) The probability of $X = 3$ is the largest probability.

D) Across repeated sets of trials, X typically varies by about 3 successes from its mean of 12.

APSTAT-MCQ5-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A test of $H_0: p = 0.50$ has power 0.85 when the true population proportion is $p = 0.60$. What does this mean?

- A) If $p = 0.50$, the test rejects H_0 in 85% of repeated samples.
- B) After one rejection, there is an 85% probability that $p = 0.60$.
- C) If $p = 0.60$, the test will reject H_0 in about 85% of repeated samples of the planned size.
- D) The test's Type I error probability is 0.15.

APSTAT-MCQ5-040

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Researchers want to estimate the mean depth of a reservoir. Which plan is most likely to avoid the selection bias of measuring only near the shoreline?

- A) Measure at the 100 easiest shoreline locations.
- B) Overlay a grid, randomly select coordinates across the entire water surface, and measure depth at those locations.
- C) Ask boaters to submit depths from locations they choose.
- D) Measure along one path from the boat ramp and treat all readings as an SRS.

APSTAT-MCQ5-041

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A 95% confidence interval for a population mean length is (1.42, 1.58) meters. If lengths are reported in centimeters instead, which interval and confidence level are correct?

- A) (142, 158) centimeters at 95% confidence

- B) (0.0142, 0.0158) centimeters at 95% confidence
- C) (142, 158) centimeters at 0.95% confidence
- D) (1.42, 1.58) centimeters at 95% confidence

APSTAT-MCQ5-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Two quantitative datasets have exactly the same five-number summary. Which statement must be true?

- A) Their histograms have the same number of modes.
- B) Their means and standard deviations are equal.
- C) Their individual data values are identical.
- D) Their boxplots can be identical even though gaps, clusters, and other internal patterns differ.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ5-001, Correct: A, Correct Answer: -5 , Explanation: Deviations from the sample mean sum to zero. The five known deviations sum to $-4 - 1 + 2 + 3 + 5 = 5$, so k must be -5 .

APSTAT-MCQ5-002, Correct: C, Correct Answer: The tiny p -value does not repair possible voluntary-response and undercoverage bias in the poll., Explanation: A p -value measures sampling evidence under a probability model; it cannot correct a nonrepresentative data-collection process. A very large opt-in sample can still be systematically biased.

APSTAT-MCQ5-003, Correct: D, Correct Answer: A one-sample t -test for a population mean, Explanation: One random sample provides a quantitative response to be compared with one claimed population mean. No second group or paired condition is present.

APSTAT-MCQ5-004, Correct: B, Correct Answer: -0.70 , Explanation: For n paired observations, $r = \frac{\sum(z_x z_y)}{(n - 1)}$. Therefore $r = \frac{-3.5}{(6 - 1)} = \frac{-3.5}{5} = -0.70$.

APSTAT-MCQ5-005, Correct: D, Correct Answer: 0.10 , Explanation: Probabilities must sum to 1, so $c[(0 + 1) + (1 + 1) + (2 + 1) + (3 + 1)] = 10c = 1$. Thus $c = 0.10$.

APSTAT-MCQ5-006, Correct: A, Correct Answer: 0.70 , Explanation: With $c = 0.10$, $P(X = 2) = 0.30$ and $P(X = 3) = 0.40$. Therefore $P(X \geq 2) = 0.30 + 0.40 = 0.70$.

APSTAT-MCQ5-007, Correct: B, Correct Answer: $X = 3$ is most likely, with probability 0.40 , but an individual installation is not guaranteed to have 3 features., Explanation: The probabilities for $x = 0, 1, 2, 3$ are $0.10, 0.20, 0.30$, and 0.40 . The mode is 3, but a probability of 0.40 is neither certainty nor an exact short-run frequency.

APSTAT-MCQ5-008, Correct: C, Correct Answer: The sampling interval matches the production cycle, so the sample may repeatedly observe the same cycle position., Explanation: Periodicity matching the interval can make the systematic sample unrepresentative, even with a random start, because only one position in each cycle is repeatedly inspected.

APSTAT-MCQ5-009, Correct: A, Correct Answer: 10, Explanation: When each true null is tested at $\alpha = 0.05$, the expected number of false rejections among 200 independent tests is $200(0.05) = 10$.

APSTAT-MCQ5-010, Correct: D, Correct Answer: $\bar{x} = 53.422$, Explanation: Rejection begins when $t = (\bar{x} - 50)/2 = 1.711$. Thus $\bar{x} = 50 + 1.711(2) = 53.422$.

APSTAT-MCQ5-011, Correct: B, Correct Answer: Randomly select fast-charged batteries and record the proportion that overheat., Explanation: Conditioning on fast charging restricts the reference group to fast-charged batteries. The overheating proportion within that group estimates $P(\text{overheats} | \text{fast-charged})$.

APSTAT-MCQ5-012, Correct: C, Correct Answer: 0.20, Explanation: In a density histogram, proportion is represented by area, not height alone. The bar has width 5 and height 0.04, so its area is $5(0.04) = 0.20$.

APSTAT-MCQ5-013, Correct: C, Correct Answer: How is engine temperature associated with fuel efficiency across a sample of vehicles?, Explanation: Regression analyzes the relationship between two quantitative variables measured on the same observational units. The other questions concern proportions, mean comparisons, or categorical distributions.

APSTAT-MCQ5-014, Correct: A, Correct Answer: A probability sample of residential addresses, including households without listed telephone numbers, Explanation: An address-based frame can include households omitted from a published telephone directory. Increasing a biased directory sample does not repair undercoverage, and voluntary or store-based samples introduce other selection biases.

APSTAT-MCQ5-015, Correct: B, Correct Answer: (0.96, 8.24) minutes, Explanation: The margin of error is $2.024(1.8) = 3.643$ minutes. Thus 4.6 ± 3.643 gives approximately (0.96, 8.24) minutes.

APSTAT-MCQ5-016, Correct: D, Correct Answer: 78th percentile, Explanation: The proportion of scores at or below this score is $28/36 \approx 0.778$, placing it at approximately the 78th percentile under the stated convention.

APSTAT-MCQ5-017, Correct: A, Correct Answer: $\mu_Y = \$91.40$ and $\sigma_Y = \$21.60$, Explanation: For $Y = aX + b$, $\mu_Y = a\mu_X + b$ and $\sigma_Y = |a|\sigma_X$. Thus $\mu_Y = 1.08(80) + 5 = \$91.40$ and $\sigma_Y = 1.08(20) = \$21.60$; the fixed fee does not affect spread.

APSTAT-MCQ5-018, Correct: C, Correct Answer: The population proportion p of remote employees who report isolation, Explanation: The population is employees meeting the remote-work condition, and the parameter is the isolation proportion within that population. A sample proportion is a statistic, and reversing the condition answers a different question.

APSTAT-MCQ5-019, Correct: B, Correct Answer: Divide the preserve by habitat and randomly sample locations within every habitat., Explanation: Stratifying by habitat guarantees representation of each distinct environment and can improve precision when soil moisture differs strongly among habitats.

APSTAT-MCQ5-020, Correct: D, Correct Answer: The sample mean typically varies by about 0.8 milligram from sample to sample around μ ., Explanation: Standard error describes the sampling variability of an estimator. It is not the spread of individual observations and does not give a percentage above the mean.

APSTAT-MCQ5-021, Correct: C, Correct Answer: 79.75%, Explanation: The explained proportion is $r^2 = (-0.45)^2 = 0.2025$, or 20.25%. The unexplained percentage is $100\% - 20.25\% = 79.75\%$.

APSTAT-MCQ5-022, Correct: A, Correct Answer: Among expedited orders, what proportion are delivered late?, Explanation: The condition after the vertical bar defines the reference group. $P(\text{late} \mid \text{expedited})$ is the late-delivery proportion among expedited orders, not the reverse conditional or the joint probability.

APSTAT-MCQ5-023, Correct: D, Correct Answer: The sample has fewer than 10 successes., Explanation: A standard one-proportion z-interval requires at least 10 successes and 10 failures. The sample has only 2 successes, so the large-count condition fails.

APSTAT-MCQ5-024, Correct: B, Correct Answer: Each corresponding bin contains the same proportion of its laboratory's sample, though the raw bin counts can differ., Explanation: Identical relative-frequency histograms match bin proportions and overall binned shape. Because the sample sizes differ, the raw counts need not match, and identical bins do not determine exact means, standard deviations, or individual values.

APSTAT-MCQ5-025, Correct: A, Correct Answer: Randomly rotate the sensors between the two sites across measurement periods and record sensor and site for each reading., Explanation: Rotation prevents one sensor's calibration tendency from being permanently linked to one site. Recording both factors permits the site comparison to be separated from sensor effects.

APSTAT-MCQ5-026, Correct: C, Correct Answer: 0.17, Explanation: Because the two reported causes are mutually exclusive, their intersection is zero. Therefore $P(\text{weather or mechanical}) = 0.12 + 0.05 = 0.17$.

APSTAT-MCQ5-027, Correct: B, Correct Answer: The association may be generalized to the sampled population if the sampling design was sound, but causation is not established., Explanation: Random sampling can support population generalization, while the absence of random assignment leaves possible confounding and prevents a causal conclusion.

APSTAT-MCQ5-028, Correct: D, Correct Answer: The number of minutes the participant actively edited the shared document during the session, Explanation: This definition converts the vague concept into a specific quantitative measurement for each participant. The alternatives are subjective, categorical, or measured at the group rather than participant level.

APSTAT-MCQ5-029, Correct: C, Correct Answer: 0.04, Explanation: A confidence interval is centered at its point estimate. The midpoint is $[-0.04 + 0.12]/2 = 0.04$.

APSTAT-MCQ5-030, Correct: A, Correct Answer: $\hat{y} = 38.9 - 0.38u$, Explanation: Since $u = x - 15$, $x = u + 15$. Substitution gives $\hat{y} = 44.6 - 0.38(u + 15) = 44.6 - 5.7 - 0.38u = 38.9 - 0.38u$.

APSTAT-MCQ5-031, Correct: D, Correct Answer: It is the predicted surface temperature, 38.9°C, for a roof with soil depth 15 centimeters., Explanation: The centered intercept occurs at $u = 0$, which corresponds to $x = 15$. Centering makes the intercept a prediction at a meaningful observed soil depth.

APSTAT-MCQ5-032, Correct: B, Correct Answer: The residuals sum to zero, so positive and negative prediction errors balance algebraically., Explanation: For ordinary least squares with an intercept, the fitted residuals have mean zero and therefore sum to zero, apart from rounding. Individual residuals need not be zero or equal in magnitude.

APSTAT-MCQ5-033, Correct: A, Correct Answer: To identify ambiguous wording and response-option problems before the main study, Explanation: A pilot test is used to improve the survey instrument and administration. A small nonprobability pilot does not replace the main probability sample or guarantee perfect response.

APSTAT-MCQ5-034, Correct: C, Correct Answer: 0.0103, Explanation: The approximate standard deviation is $\sqrt{[\hat{p}(1 - \hat{p})/n]} = \sqrt{[0.31(0.69)/2000]} \approx \sqrt{0.00010695} \approx 0.0103$.

APSTAT-MCQ5-035, Correct: D, Correct Answer: Fail to reject H_0 because $|1.90| < 2.086$., Explanation: A two-sided test rejects only when the statistic lies beyond either critical value. Since 1.90 is inside $(-2.086, 2.086)$, the result is not significant at $\alpha = 0.05$.

APSTAT-MCQ5-036, Correct: B, Correct Answer: 200 observations in each sample, Explanation: When the proportions are expected to be similar and the total is fixed, the variance term is minimized by balancing the sample sizes, which minimizes $1/n_1 + 1/n_2$.

APSTAT-MCQ5-037, Correct: A, Correct Answer: Mean 0 and standard deviation 1, Explanation: Subtracting $\bar{x}$ centers the transformed data at zero, and dividing by the sample standard deviation rescales the sample standard deviation to one.

APSTAT-MCQ5-038, Correct: D, Correct Answer: Across repeated sets of trials, X typically varies by about 3 successes from its mean of 12., Explanation: The standard deviation of a random variable describes typical distance from its mean across repetitions. It is not a maximum distance and does not guarantee a 68% interval without an appropriate shape condition.

APSTAT-MCQ5-039, Correct: C, Correct Answer: If $p = 0.60$, the test will reject H_0 in about 85% of repeated samples of the planned size., Explanation: Power is the probability of rejecting H_0 when a specified alternative value is true. It is not a posterior probability for p or the Type I error rate.

APSTAT-MCQ5-040, Correct: B, Correct Answer: Overlay a grid, randomly select coordinates across the entire water surface, and measure depth at those locations., Explanation: Random coordinates give locations across the reservoir a defined chance of selection and avoid restricting measurements to accessible shoreline areas.

APSTAT-MCQ5-041, Correct: A, Correct Answer: (142, 158) centimeters at 95% confidence, Explanation: Converting meters to centimeters multiplies both endpoints by 100 but does not change the confidence level or the inferential meaning.

APSTAT-MCQ5-042, Correct: D, Correct Answer: Their boxplots can be identical even though gaps, clusters, and other internal patterns differ., Explanation: A boxplot displays only the five-number

summary and possible outlier information. Datasets with the same summary can have different individual values, means, standard deviations, gaps, or modes.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ5

- APSTAT-MCQ5-001 to 005: Unit 1 temperature deviations use the zero-sum property around $\bar{x}$ to recover a missing deviation; Unit 3 a 40,000-response opt-in zoning poll shows that a tiny p-value cannot repair selection bias; Unit 4 rechargeable-lamp operating time selects a one-sample t-test against a benchmark; Unit 5 computes correlation from $\Sigma(z_{xz_y})/(n - 1)$; Unit 2 smart-home security-feature distribution defined by $P(X = x) = c(x + 1)$ solves the normalizing constant. Misconceptions include failing to balance deviations, treating sample size as a cure for voluntary response, choosing a two-sample procedure for one group, omitting $n - 1$ in the correlation formula, and normalizing probabilities incorrectly.
- APSTAT-MCQ5-006 to 010: Unit 2 formula-defined security-feature distribution sums upper point probabilities and identifies its mode without short-run certainty; Unit 1 bottling-line systematic sample detects periodicity when the interval equals a 12-position cycle; Unit 3 200 true-null tests at $\alpha = 0.05$ use expected false rejections 200α ; Unit 4 solves a one-sample t rejection boundary backward for the threshold $\bar{x}$. Misconceptions include reading a tail event from one point, treating a modal count as guaranteed, assuming a random start defeats periodicity, confusing α with power, and multiplying rather than adding the critical standard-error displacement.
- APSTAT-MCQ5-011 to 015: Unit 2 battery overheating conditional on fast charging samples from the conditioned group; Unit 1 density-histogram bar area, not height, gives a proportion; Unit 5 vehicle engine temperature and fuel efficiency identify a two-quantitative-variable regression question; Unit 3 emergency-kit prevalence replaces a telephone-directory frame with address-based probability sampling; Unit 4 coating drying times construct a 95% independent two-mean t-interval from estimate, SE, and t^* . Misconceptions include reversing a conditional, treating density height as frequency, using regression for categorical or mean-comparison questions, fixing undercoverage by enlarging the same frame, and omitting the critical multiplier.
- APSTAT-MCQ5-016 to 020: Unit 1 ordered exam-score rank converts 28 of 36 at-or-below observations to an approximate percentile; Unit 2 affine service-charge transformation applies $\mu_Y = a\mu_X + b$ and $\sigma_Y = |a|\sigma_X$; Unit 3 remote-work isolation question defines a population conditional proportion rather than a joint, reverse-conditional, or sample statistic; Unit 1 soil-moisture sampling stratifies a preserve by meadow, woodland, and marsh; Unit 4 interprets standard error as sample-to-sample variation of $\bar{x}$. Misconceptions include reversing percentile direction, adding a fixed fee to standard deviation, reversing the conditioned population, sampling one habitat only, and confusing estimator variability with individual spread.
- APSTAT-MCQ5-021 to 025: Unit 5 seedling height versus salinity calculates unexplained variation as $1 - r^2$; Unit 2 expedited-order notation translates $P(\text{late} \mid \text{expedited})$ into its conditioned reference group; Unit 3 paper-receipt data reject a standard one-proportion interval because only 2 successes occur; Unit 1 particle-diameter samples of sizes 80 and 200 compare identical relative-frequency histograms without equating raw counts or exact summaries; Unit 4 site-temperature study rotates two differently calibrated sensors to prevent sensor-site confounding. Misconceptions include reporting r or r^2 as unexplained percentage, reversing conditional probability, treating $n > 30$ as sufficient for proportion inference, confusing relative frequency with count, and permanently assigning one biased instrument per site.
- APSTAT-MCQ5-026 to 030: Unit 2 mutually exclusive weather and mechanical flight delays use the addition rule with zero intersection; Unit 3 random-sample vegetarian-diet and supplement-use association supports generalization but not causation; Unit 1 collaborative-writing engagement receives a participant-level quantitative operational definition; Unit 3 midpoint of a $p_A - p_B$ interval recovers the observed point estimate; Unit 5 centered green-roof soil depth begins by rewriting the fitted

temperature equation with $u = x - 15$. Misconceptions include multiplying disjoint probabilities, equating random sampling with random assignment, accepting subjective or group-level definitions, confusing interval width with center, and changing the slope when centering x .

- APSTAT-MCQ5-031 to 035: Unit 5 centered green-roof regression interprets the new intercept at $x = 15$ and uses the zero-sum residual property; Unit 1 questionnaire pilot cognitive interviews identify wording and response-option problems; Unit 2 simulation proportion $\hat{p} = 0.31$ from 2,000 trials calculates approximate Monte Carlo standard deviation; Unit 4 two-sided $t = 1.90$ compares with critical values ± 2.086 . Misconceptions include interpreting the centered intercept at $x = 0$, assuming every residual is zero, treating a pilot as the final probability sample, omitting $\sqrt{n}$ from simulation variability, and comparing t directly with α .

- APSTAT-MCQ5-036 to 040: Unit 3 fixed-total two-proportion sampling minimizes standard error with balanced n_1 and n_2 ; Unit 1 sample z -scores have mean 0 and sample standard deviation 1; Unit 2 binomial $\sigma_X = 3$ is interpreted as typical count variability around $\mu_X = 12$; Unit 3 power 0.85 at $p = 0.60$ receives a repeated-sampling interpretation; Unit 1 reservoir-depth sampling uses random grid coordinates over the whole surface to avoid shoreline selection bias. Misconceptions include assuming allocation is irrelevant, retaining original center or spread after standardizing, treating σ as a maximum distance, interpreting power as a posterior probability, and treating convenient shoreline measurements as random.

- APSTAT-MCQ5-041 to 042: Unit 4 converts a 95% mean interval from meters to centimeters without changing confidence; Unit 1 identical five-number summaries can yield identical boxplots despite different gaps, clusters, means, standard deviations, and raw values. Misconceptions include changing confidence during unit conversion and assuming a boxplot uniquely determines a distribution.

- Unit counts for APSTAT-MCQ5: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: formula-defined smart-home security-feature distribution, APSTAT-MCQ5-005 to 007; centered green-roof soil-depth regression, APSTAT-MCQ5-030 to 032.